



UNSW
THE UNIVERSITY OF NEW SOUTH WALES



Lecture Week 1: Introduction to Bayesian inference and Monte Carlo

Lecturer: Prof. Scott Sisson

Outline

1. Probability Basics & Frequentist Statistics
2. What is Bayesian statistics?
 - Bayes formula, prior and posterior probabilities
3. Monte Carlo integration basics
 - What is this, and why do we need it?

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What is Frequentist Statistics?



Example: Forestry commission wish to estimate proportion of trees, $\theta \in [0, 1]$, in a large forest with a particular disease.

Select a sample of n trees.

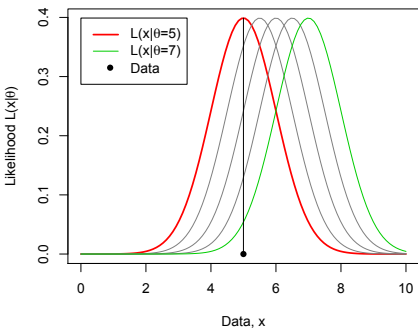
Specify probability model: $L(x|\theta) :$
 $X|\theta \sim \text{Binomial}(n, \theta)$

maximise likelihood of $X = x$ w.r.t. θ .

Frequentist inference in the form of

- ▶ a **point estimate** ($\hat{\theta} = 0.1$);
- ▶ a **confidence interval** (95% CI for θ is (0.08, 1.20));
- ▶ a **hypothesis test** (reject $H_0 : \theta = 0.07$ at 5% level);
- ▶ a **prediction** (15% of trees will be infected next year);
- ▶ a **decision** (how many infected trees to remove).

Frequentist statistics recap: Maximum Likelihood



Likelihood function: $L(x|\theta)$

- ▶ x is the random variable (**not θ !**)
- ▶ θ is fixed
- ▶ θ is considered as a constant

MLE: $\hat{\theta} = \arg_{\theta} \max L(x|\theta)$

- ▶ Change θ so that x is given largest likelihood
- ▶ Treat as a function of θ (**but it isn't**)

MLE properties

- ▶ In many circumstances $\hat{\theta} \sim N(\theta, H)$
- ▶ Allows construction of confidence intervals for $\hat{\theta}$ and hypothesis tests

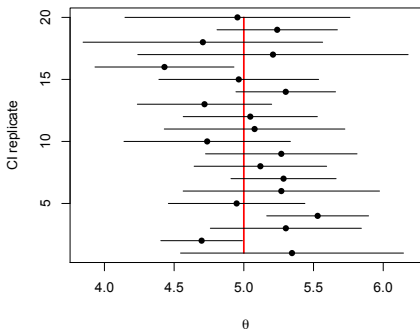
Model Prediction

- ▶ Distribution of future data y obtained via $y \sim \pi(y|\hat{\theta})$

Frequentist Statistics: Confidence Intervals

$100(1 - \alpha)\%$ CI for pop. mean (θ):

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$



Interpretation:

- ▶ Very awkward!
- ▶ Would like to say e.g.:
"there is a 95% probability that θ is in your (single) confidence interval"
- ▶ But we can't

Our (single) CI either contains θ ($\text{Pr} = 1$) or it doesn't ($\text{Pr} = 0$).

Actual interpretation: in the long run over all experiments/datasets 95% of CIs will contain the true parameter

Frequentist Statistics: Non-regular likelihoods

Hospital Example:

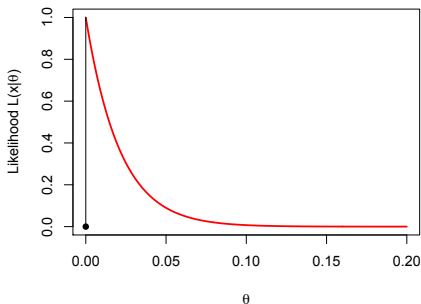
θ = Pr(death of baby from operation)

r = Number of previous deaths

n = Number of previous operations

Observed data: $r = 0$, $n = 47$

$$r \sim \text{Binomial}(n, \theta)$$



The likelihood:

- ▶ $\hat{\theta} = 0$ is still well defined
- ▶ But $\hat{\theta} \sim N(\theta, H)$ fails
- ▶ How to get a CI for θ ?
- ▶ How to do hypothesis test?

Standard likelihood asymptotics
can fail even in simple situations

Need non-standard likelihood
asymptotics – can be tricky.

Frequentist Statistics: Predictions

Hospital Example:

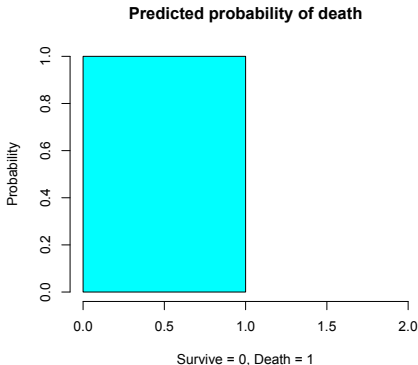
Predictive distribution is

$$\begin{aligned} y &\sim \pi(y|\hat{\theta}) \\ &= \text{Bernoulli}(\hat{\theta}) \end{aligned}$$

with $\hat{\theta} = 0$.

Predictive distribution will produce:

- ▶ 100% prediction for survival
- ▶ 0% prediction for death



This is completely unrealistic

Performing model predictions using a fixed point estimate, $\hat{\theta}$, can cause problems

Does not take **uncertainty about θ** into account

Can use profile likelihoods – but is there an alternative approach?

Probability Basics

Conditional Probability:

$$Pr(A|B) = \frac{Pr(A, B)}{Pr(B)}$$

Law of total probability (discrete r.v. B)

$$Pr(A) = \sum_i Pr(A, B_i) = \sum_i Pr(A|B_i)Pr(B_i)$$

Law of total probability (continuous r.v. X) (a.k.a marginalisation)

$$Pr(A) = \int Pr(A|X = x)p(x)dx = \int Pr(A, x)dx$$

$p(x)$ = pdf of X .

Bayes Theorem (conditional probability-type statement)

$$Pr(A|B) = \frac{Pr(B|A)Pr(A)}{Pr(B)}$$

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What is Bayesian Statistics?

A more general question: how do we optimally *learn*?

- ▶ θ – an object (parameter?) that we wish to learn about
- ▶ $\pi(\theta)$ – current knowledge about θ (a distribution?)
- ▶ x – information (data?) we observe that can tell us more about θ
- ▶ How to update our knowledge from $\pi(\theta)$ to $\pi(\theta|x)$?
- ▶ Want to identify function ψ such that:

$$\pi(\theta|x) = \psi(x, \pi(\theta))$$

- ▶ Two options:
 1. Simple option: Bayes' Theorem (this lecture)
 2. Modern lens: Optimisation-centric updating (see later)

What is Bayesian Statistics?



T. Bayes.

Wikipedia page:

http://en.wikipedia.org/wiki/Thomas_Bayes

Thomas Bayes (1701—1761)

- ▶ English mathematician and Presbyterian minister
- ▶ 1763 publication:
An Essay towards solving a Problem in the Doctrine of Chances

Contained a special case of what is now known as **Bayes' Theorem**:

$$Pr(B|A) = \frac{Pr(A|B)Pr(B)}{Pr(A)}$$

- ▶ Essentially a conditional probability statement

What is Bayesian Statistics?

Where does Bayes' formula come from?

$$\left. \begin{aligned} Pr(A \cap B) &= Pr(A | B)Pr(B) \\ Pr(A \cap B) &= Pr(B | A)Pr(A) \end{aligned} \right\} \Rightarrow \text{Equate}$$

$$Pr(B|A) = \frac{Pr(A|B)Pr(B)}{Pr(A)}$$

Conceptual differences between Frequentist & Bayesian:

Frequentist:

Data x is **random** (e.g. $x \sim N(\theta, 1)$).

Parameter θ (though unknown) is a **constant**.

Bayesian:

Data x is **random**, as before.

Parameter θ (though unknown) is **also random** (i.e. $\theta \sim \pi(\theta)$).

Allows easy descriptive inference in terms of direct probabilities.

How does Bayes Theorem help us do this?

If we set:

$A = x$ = observed data

$B = \theta$ = unknown parameters

Then:

$$\begin{aligned}Pr(\theta|x) &= \frac{Pr(x|\theta)Pr(\theta)}{Pr(x)} \\ \pi(\theta|x) &= \frac{L(x|\theta)\pi(\theta)}{m(x)} = \frac{L(x|\theta)\pi(\theta)}{\int_{\Theta} L(x|\theta)\pi(\theta)d\theta} \\ \pi(\theta|x) &\propto L(x|\theta)\pi(\theta) \quad (m(x) \text{ not a function of } \theta)\end{aligned}$$

Here:

$\pi(\theta)$ = Prior distribution

$L(x|\theta)$ = Likelihood function – assumes a parameterised model

$m(x)$ = Normalisation constant

$\pi(\theta|x)$ = Posterior distribution

Prior $\rightarrow$ Posterior

$$\pi(\theta|x) = \psi(x, \pi(\theta)) = \frac{L(x|\theta)\pi(\theta)}{m(x)}$$

$\pi(\theta)$ = Prior distribution:

Describes our knowledge about the model parameters:

- ▶ in distributional form
- ▶ before we have seen the data.

Can be “uninformative” (e.g. “flat”) or involve expert elicitation

$\pi(\theta|x)$ = Posterior distribution:

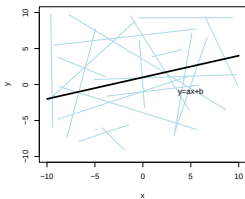
Describes our knowledge about the model parameters:

- ▶ in distributional form
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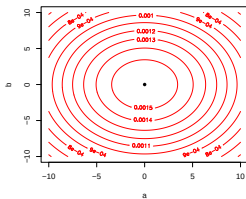
Going from $\pi(\theta) \rightarrow \pi(\theta|x)$ reflects how our beliefs about the model parameters, θ , have changed by observing the data, x .

Illustrative idea (regression)

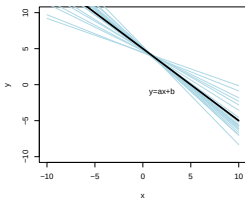
Regression line: $y=ax+b$



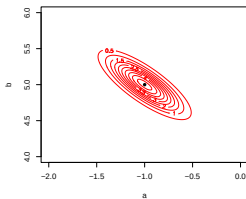
Prior distribution for (a,b)



Regression line: $y=ax+b$



Posterior distribution for (a,b)



Fitting a linear regression:

$$y = ax + b + \epsilon$$

Parameters: $\theta = (a, b)'$

Prior information about (a, b) :
 $\pi(a, b)$, before seeing data, in
distributional form e.g.

$$\pi \left(\begin{pmatrix} a \\ b \end{pmatrix} \right) = N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \right)$$

After fitting the model to data
prior $\rightarrow$ posterior belief about
 (a, b) . Still in distributional
form

$$\pi \left(\begin{pmatrix} a \\ b \end{pmatrix} \mid x \right) = N(\hat{\mu}, \hat{\Sigma})$$

Key point:

Parameters are represented by distributions, not point estimates

Do we really have prior knowledge?

Example #1: Previous regression example

We specified prior information about (a,b) : $\pi(a,b)$, before seeing data, as

$$\pi \begin{pmatrix} a \\ b \end{pmatrix} = N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \right)$$

Before seeing the data x :

- ▶ Normal distribution: Credible (but plenty of alternative distributions that could be better)
- ▶ Zero mean: Credible (equally likely to see $+/ -$ intercept & slope)
- ▶ Equal variances: Likely not (consider $\text{Var}(\hat{\beta})$ for $\hat{\beta} = (X'X)^{-1}X'Y$), but probably not a big deal
- ▶ Zero covariance: Probably not (expect this to be negative)

Often, sensible prior information can be obtained just from the model (which was also specified with prior knowledge!)

Do we really have prior knowledge?

Example #2:

We have a model $X|\theta \sim \text{Binomial}(10, \theta)$ and observe $x = 10$.

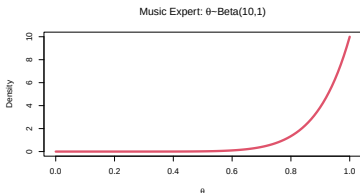
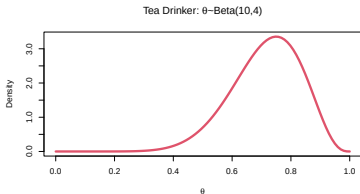
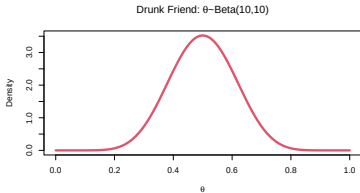
Hypothesis $H_0 : \theta \leq 0.5$ is rejected in favour of $H_1 : \theta > 0.5$ each time.

3 different scenarios (same data & model):

- ▶ A drunk friend claims she can predict the outcome of tossing a fair coin, and does so correctly for 10 tosses.
- ▶ A tea-drinker claims she can detect whether milk was added before or after the tea, just by taste. She does so correctly for 10 cups.
- ▶ A music expert claims he can distinguish between a page of Hayden's and Mozart's work. He correctly categorises 10 pieces.

In terms of data only, we must draw the same conclusion in each case.

Do we really have prior knowledge?



But our **prior beliefs** suggest:

- ▶ being highly skeptical about the drunk friend: $\pi(\theta) = \text{Beta}(10, 10)$
- ▶ slightly impressed about the tea-drinker: $\pi(\theta) = \text{Beta}(10, 4)$
- ▶ not at all surprised about the music expert: $\pi(\theta) = \text{Beta}(10, 1)$

Bayesian validity

Argument for priors:

It is sensible to include prior information together with inference, as we invariably have some knowledge about the model parameters.

Argument against priors:

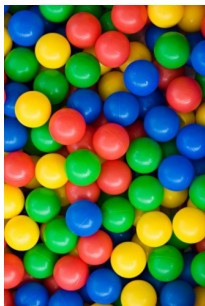
Different prior beliefs lead to different inferences.

Whether you see the argument against priors as a good or bad thing will determine your acceptability of the Bayesian approach.

Things to bear in mind:

- ▶ Prior knowledge is used to construct a likelihood model;
- ▶ Prior knowledge is used to determine test significance levels (can choose a high level to ensure rejection of “unlikely” hypothesis);
- ▶ Frequentist statistics maximises the likelihood surface. Bayesian statistics averages over it (see later). The prior just weights the averaging. Weighting is completely uncontroversial in Frequentist statistics (think weighted regression).

Example: Coloured balls



Frequentist problem:

In a bag there are 6 balls: 3 red and 3 black. Three balls are drawn without replacement. Find the probability that no black ball is left in the bag.

Standard probability calculations. E.g.

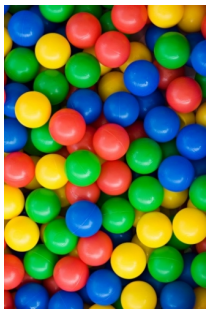
$$P = \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} = \frac{1}{20}$$

Alternative question:

In a bag there are 6 balls of unknown colour. Three balls are drawn without replacement and are found to be black. Find the probability that no black ball is left in the bag.

How to solve?

Example: Coloured balls



In a bag there are 6 balls of unknown colour. Three balls are drawn without replacement and are found to be black. Find the probability that no black ball is left in the bag.

Bayes solution:

- ▶ A = Event (3 black balls are drawn)
- ▶ C_i = Event (there were i black balls in the bag).

Then by Bayes' theorem

$$\Pr(C_3|A) = \frac{\Pr(A|C_3)\Pr(C_3)}{\sum_{i=0}^6 \Pr(A|C_i)\Pr(C_i)}$$

- ▶ But what values to give to $\Pr(C_0), \dots, \Pr(C_6)$?
- ▶ The prior probabilities of the number of black balls in the bag, *before* we have seen the data
- ▶ Without any other information might assume $\Pr(C_0) = \dots = \Pr(C_6) = \frac{1}{7}$.

Example: Coloured balls

In this case, we can then compute

$$\begin{aligned}\Pr(C_3|A) &= \frac{\Pr(A|C_3)\Pr(C_3)}{\sum_{i=0}^6 \Pr(A|C_i)\Pr(C_i)} \\&= \frac{(\frac{3}{6} \times \frac{2}{5} \times \frac{1}{4}) \times \frac{1}{7}}{\frac{1}{7}[0 + 0 + 0 + (\frac{3}{6} \times \frac{2}{5} \times \frac{1}{4}) + (\frac{4}{6} \times \frac{3}{5} \times \frac{2}{4}) + (\frac{5}{6} \times \frac{4}{5} \times \frac{3}{4}) + (\frac{6}{6} \times \frac{5}{5} \times \frac{4}{4})]} \\&= \frac{1}{35}.\end{aligned}$$



- ▶ Prior of $\Pr(C_3) = \frac{1}{7}$ has been updated to $\Pr(C_3|A) = \frac{1}{35}$
- ▶ Much less likely than before
- ▶ Does this make sense?

(Yes: for $\Pr(C_k|A)$, $k = 3$ is least likely event for $k \geq 3$.)

Example: Coloured balls



But is this the most sensible prior?

Suppose you know that **the ball manufacturer make 10 different colours.**

You might take a prior view that each ball is black with probability $\frac{1}{10}$ as the basis for calculating $\Pr(C_i)$.

Or suppose that you know **these are snooker balls** (with a specific distribution of colours).

The point is, we often have to think hard about our prior beliefs, and the answer we get will depend on what we believe.

(More on priors next week, including “uninformative” priors.)



Summary: Bayesian updating

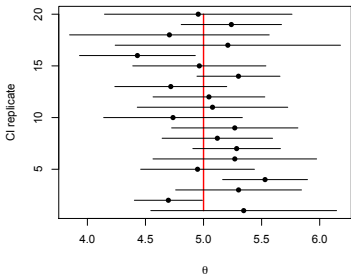
There are 4 key steps in the Bayesian approach:

- ▶ Specification of a likelihood model $L(x|\theta)$
- ▶ Determination/elicitation of a suitable prior distribution $\pi(\theta)$
- ▶ Calculation of the posterior distribution via Bayes' Theorem

$$\pi(\theta|x) = \frac{L(x|\theta)\pi(\theta)}{\int_{\theta} L(x|\theta)\pi(\theta)d\theta}$$

- ▶ Draw inference (and/or make decisions) from the posterior.

Confidence vs Credible Intervals



Frequentist Interpretation (θ is fixed):

"In the long run over many experiments, 95% of CI's will contain the true parameter"

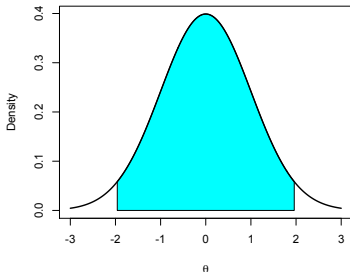
(But your single CI either will or won't)

Bayesian Interpretation (θ is random):

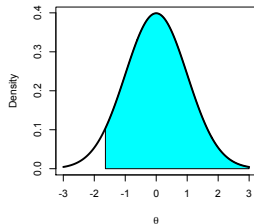
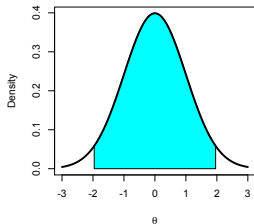
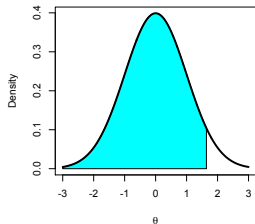
"There is a 95% probability that θ is in your (single) credible interval"

Very intuitive and simple.

Credible versus Confidence Interval



A comment on Credible Intervals



Credible intervals are not unique – there are infinitely many!

All have exactly the same interpretation (the choice is yours).

High Density Region (HDR) intervals:

- ▶ The credible interval with the **shortest width (or smallest area)**
- ▶ Not always the central interval e.g. if posterior is skewed
- ▶ Can also comprise **disjoint regions**

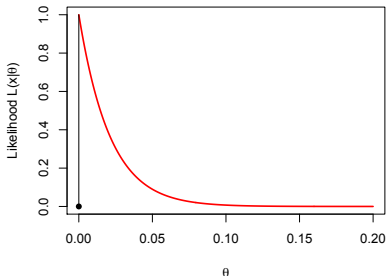
Non-regular likelihoods

Observed data: $r = 0$, $n = 47$

$$r \sim \text{Binomial}(n, \theta)$$

The likelihood:

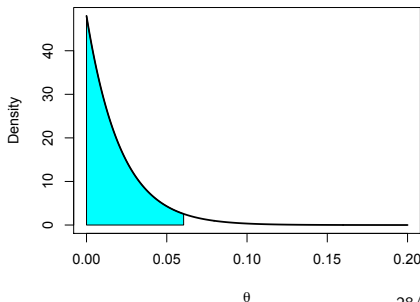
- ▶ $\hat{\theta} = 0$ is still well defined
- ▶ But $\hat{\theta} \sim N(\theta, H)$ fails
- ▶ How to get a CI for θ ?



The Bayesian approach

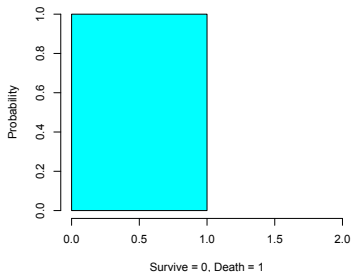
- ▶ $\theta \sim \pi(\theta) = U(0, 1)$ prior
- ▶ No requirement of “standard” likelihood asymptotics.
- ▶ The posterior contains all required information.

Posterior distribution $\pi(\theta|x)$



Predictive distributions

Predicted probability of death



Hospital Example:

Classical predictive distribution will produce:

- ▶ 100% prediction for survival
- ▶ 0% prediction for death

This is completely unrealistic

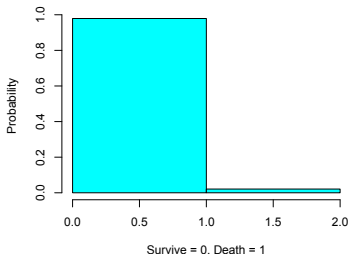
Bayesian predictive procedure:

- ▶ $\theta \sim \pi(\theta|x)$ draw from posterior
- ▶ $y \sim \text{Bernoulli}(\theta)$

At least **some** predicted possibility of future deaths

Accounts for parameter uncertainty.

(Bayesian) Predicted probability of death



How to compute the posterior distribution?



There are two practical ways of deriving the posterior distribution

1) Algebraically (least common):

- ▶ All calculations are exact
- ▶ Calculations can be difficult
- ▶ Not always possible to obtain exact results (integrations only possible in restricted cases)

2) Computationally (most common):

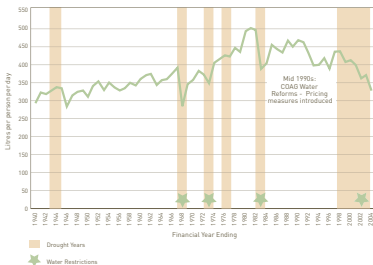
- ▶ All calculations are approximate
- ▶ Calculations are much simpler
- ▶ Can always obtain results

We cover both options in this course. Why?

- ▶ Exact solutions save your CPU/GPU cycles
- ▶ Essential to know the maths behind the methods you implement(!)

Example: Water Consumption

Average daily per capita water use⁴
Melbourne 1940-2004*



* NOTE: Figure for 2003-04 is forecasted estimation

Melbourne average daily per capita water use (litres) from 1940-2004

Data is discrete, so one possible model is $x_i \sim \text{Poisson}(\theta)$, $i = 1, \dots, n$.

Independence, and constant rate θ is clearly wrong, but it simplifies things for now (we will return to this example later).

$$\sum_{i=1}^n x_i = 24890, n = 65$$

Year	1940	1941	1942	1943	1944	1945	1946	...
Litres (x_i)	300	320	315	330	340	335	290	...

Example: Water Consumption

$L(x|\theta)$ = Likelihood function:

$$x_i \sim \text{Poisson}(\theta): \quad \pi(x|\theta) = \frac{\theta^x}{x!} \exp(-\theta) \quad i = 1, \dots, n, \theta > 0.$$

$$L(x|\theta) = \prod_{i=1}^n \left(\frac{\theta^{x_i}}{x_i!} \exp(-\theta) \right) = \frac{\theta^{\sum_i x_i} \exp(-n\theta)}{\prod_i x_i!}$$

Example: Water Consumption

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$\pi(\theta)$ = Prior distribution:

$$\theta \sim \text{Gamma}(\alpha, \beta): \quad \pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta) \quad \alpha, \beta > 0$$

Example: Water Consumption

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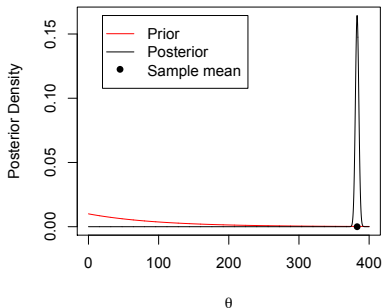
$$\theta \sim \text{Gamma}(\alpha, \beta): \quad \pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta) \quad \alpha, \beta > 0$$

$\pi(\theta|x)$ = Posterior distribution:

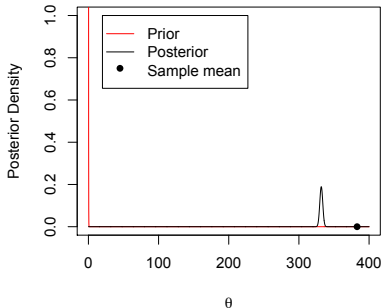
$$\begin{aligned} \pi(\theta|x) &\propto \frac{\theta^{\sum_i x_i} \exp(-n\theta)}{\prod_i x_i!} \times \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta) \\ &\propto \theta^{\sum_i x_i} \exp(-n\theta) \times \theta^{\alpha-1} \exp(-\beta\theta) \\ &= \theta^{(\alpha + \sum_i x_i) - 1} \exp(-\theta(\beta + n)) \\ &\propto \text{Gamma}\left(\alpha + \sum_{i=1}^n x_i, \beta + n\right) \end{aligned}$$

Example: Water Consumption

(a) Moderate prior



(b) Strongly informative prior



Prior specification: $\alpha = 1, \beta = 0.01$

Posterior is **directly** influenced by prior.

$\alpha = 1, \beta = 10$

Prior choice:

- ▶ Can choose “uninformative” priors (see next week)
- ▶ Better to carefully elicit (justifiable) prior information
- ▶ Enough data can overwhelm prior information.

Posterior Inference

All information needed for further inference is contained in the posterior distribution

Examples:

- Point estimate (mean) of θ :

$$\bar{\theta} = \mathbb{E}_{\pi}[\theta] = \int_{\Theta} \theta \pi(\theta|x) d\theta = \dots = \frac{\alpha + \sum_i x_i}{\beta + n}$$

- Variance of θ :

$$\begin{aligned} \text{Var}(\theta) &= \mathbb{E}_{\pi}[\theta^2] - \mathbb{E}_{\pi}[\theta]^2 = \int_{\Theta} \theta^2 \pi(\theta|x) d\theta - \left(\int_{\Theta} \theta \pi(\theta|x) d\theta \right)^2 \\ &= \dots = \frac{\alpha + \sum_i x_i}{(\beta + n)^2} \end{aligned}$$

- 95% credible interval $[a, b]$ for θ : Solve

$$\int_0^a \pi(\theta|x) d\theta = 0.025 \quad \text{and} \quad \int_0^b \pi(\theta|x) d\theta = 0.975$$

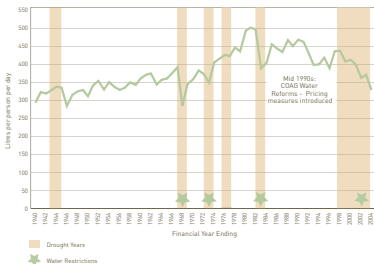
Posterior Inference

Examples:

- Predictive distribution for future data:

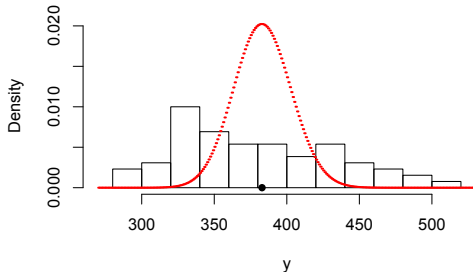
$$\begin{aligned} p(y|x) &= \int_{\Theta} \pi(y|\theta) \pi(\theta|x) d\theta \\ &= \dots = \text{NegBin} \left(y \mid \alpha + \sum_{i=1}^n x_i, \frac{\beta + n}{1 + \beta + n} \right) \end{aligned}$$

Average daily per capita water use⁴
Melbourne 1940-2004*



* NOTE: Figure for 2003-04 is forecasted estimation

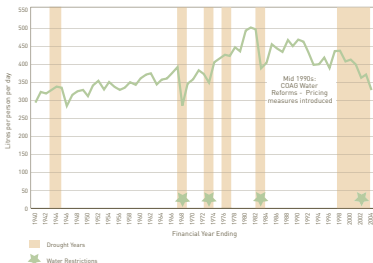
Predictive Density of y



Note: Maths possible, though laborious. **Q:** is there an easier way?
(**Answer:** Yes – see later!)

Example: Water Consumption (cont.)

Average daily per capita water use⁶
Melbourne 1940-2004*



* NOTE: Figure for 2003-04 is forecasted estimation

More realistic models

$x_i \sim \text{Poisson}(\theta)$ is clearly not realistic

Alternatives: $x_i \sim \text{Poisson}(\theta_i)$

► Linear trend:

$$\theta_i = \theta_0 + \theta_1 \text{year}_i$$
$$(\theta_0, \theta_1) \sim \pi(\theta_0, \theta_1)$$

► Piecewise linear trend (known κ):

$$\theta_i = \theta_0 + \theta_1 \text{year}_i + \theta_2 (\text{year}_i - \kappa)_+$$

e.g. $\kappa = 1982$.

$$(\theta_0, \theta_1, \theta_2) \sim \pi(\theta_0, \theta_1, \theta_2)$$

► Piecewise linear trend (unknown κ):

$$\theta_i = \theta_0 + \theta_1 \text{year}_i + \theta_2 (\text{year}_i - \kappa)_+$$
$$(\theta_0, \theta_1, \theta_2, \kappa) \sim \pi(\theta_0, \theta_1, \theta_2, \kappa)$$

Clearly it is simple to construct more realistic models.

How to do: Bayesian inference for multi-parameter models?

Posterior Inference (multivariate)

All posterior inference proceeds via integration over the posterior.
Any parameters not of interest are integrated out.

Examples: Assume posterior distribution $\pi(\theta_0, \theta_1, \theta_2, \kappa|x)$

- Marginal distribution of change-point κ :

$$\pi(\kappa|x) = \int_{\Theta_2} \int_{\Theta_1} \int_{\Theta_0} \pi(\theta_0, \theta_1, \theta_2, \kappa|x) d\theta_0 d\theta_1 d\theta_2$$

- Point estimate (mean) of θ_2 :

$$\begin{aligned}\bar{\theta}_2 &= \int_{\Theta_2} \theta_2 \pi(\theta_2|x) d\theta_2 \\ &= \int_{\Theta_2} \theta_2 \int_{\Theta_1} \int_{\Theta_0} \sum_{\kappa} \pi(\theta_0, \theta_1, \theta_2, \kappa|x) d\theta_0 d\theta_1 d\theta_2\end{aligned}$$

- 95% credible interval $[a, b]$ for θ_2 : Solve

$$\int_0^a \pi(\theta_2|x) d\theta_2 = 0.025 \quad \text{and} \quad \int_0^b \pi(\theta_2|x) d\theta_2 = 0.975$$

Posterior Inference (multivariate)

Examples:

- Predictive distribution for future data:

$$\begin{aligned} p(y|x, \text{year}(y)) &= \int_{\Theta} \pi(y|\theta) \pi(\theta|x) d\theta \\ &= \int_{\Theta_2} \int_{\Theta_1} \int_{\Theta_0} \sum_{\kappa} \pi(y|\theta_0, \theta_1, \theta_2, \kappa) \\ &\quad \times \pi(\theta_0, \theta_1, \theta_2, \kappa|x) d\theta_0 d\theta_1 d\theta_2 \\ &= \dots? \end{aligned}$$

Summary:

Procedure is same as previously, and conceptually simple.

However, **lots of difficult algebra** even for simple models.

Closed solutions typically not possible.

This is the reason **Bayesian statistics was not a viable procedure** until the early 1990s.

Outline

1. Probability Basics & Frequentist Statistics
2. What is Bayesian statistics?
 - Bayes formula, prior and posterior probabilities
3. Monte Carlo integration basics
 - What is this, and why do we need it?

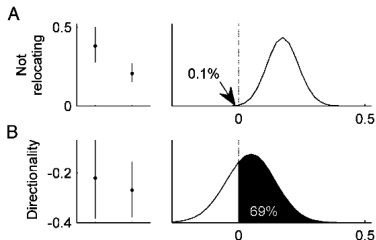
What is Monte Carlo integration?

Want to compute posterior quantities.

E.g. **Posterior mean** or $Pr(\mu_1 - \mu_2 > 0|x)$

This **requires integration of posterior**

But it **can be difficult(!)** to analytically integrate the posterior distribution to compute these.

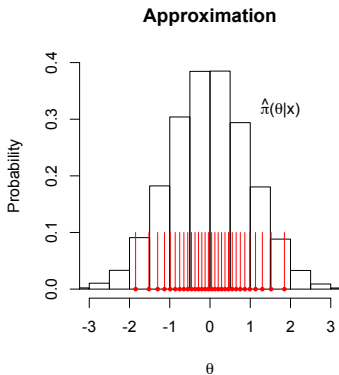
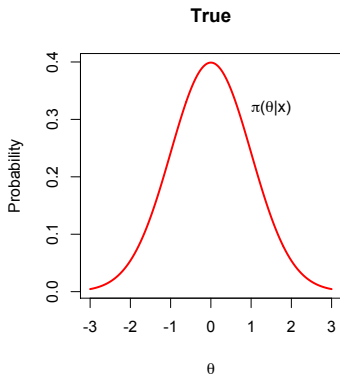


Monte Carlo refers to **randomness**.

Think of 'random' casino games of chance.
So **Monte Carlo integration** refers to **integration using random numbers**.

How does this work?

A sample approximation to the posterior



Posterior mean (exact)

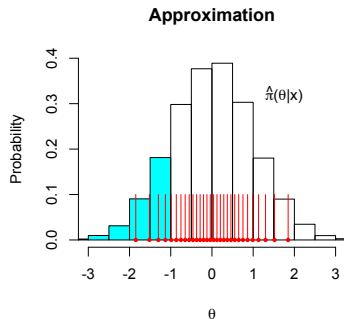
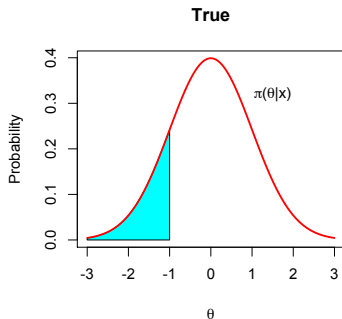
$$\begin{aligned} &= \int_{\Theta} \theta \pi(\theta|x) d\theta \\ &= \mathbb{E}_{\pi}[\theta] \end{aligned}$$

Posterior mean (approximation)

$$\approx \frac{1}{N} \sum_{i=1}^N \theta^{(i)}$$

where $\theta^{(1)}, \dots, \theta^{(N)} \sim \pi(\theta|x)$.

A sample approximation to the posterior



$$\begin{aligned}\Pr(\theta < -1) &= \\ \int_{-\infty}^{-1} \pi(\theta|x) d\theta &= \\ \int_{\Theta} I(\theta < -1) \pi(\theta|x) d\theta &= \\ \mathbb{E}_{\pi}[I(\theta < -1)]\end{aligned}$$

$$\Pr(\theta < -1) \approx \frac{1}{N} \sum_{i=1}^N I(\theta^{(i)} < -1)$$

where $\theta^{(1)}, \dots, \theta^{(N)} \sim \pi(\theta|x)$.

Why does this work?

Monte Carlo integration:

Estimate posterior quantity of interest using sample from posterior

(Heuristic) justification for expectations:

$$\pi(\theta|x) \approx \frac{1}{N} \sum_{i=1}^N \delta_{\theta^{(i)}}(\theta) \quad (\text{a discrete measure})$$

$\delta_b(A)$ is Dirac mass so that $\delta_b(A) = 1$ if $b \in A$, 0 otherwise.

Computing expectations:

$$\begin{aligned} \mathbb{E}_{\pi}[g(\theta)] &= \int_{\Theta} g(\theta) \pi(\theta|x) d\theta \\ &\approx \int_{\Theta} g(\theta) \frac{1}{N} \sum_{i=1}^N \delta_{\theta^{(i)}}(\theta) d\theta \\ &= \frac{1}{N} \sum_{i=1}^N g(\theta^{(i)}) \end{aligned}$$

Why does this work?

Asymptotics: For i.i.d $\theta^{(i)} \sim \pi(\theta|x)$,

$$\frac{1}{N} \sum_{i=1}^N g(\theta^{(i)}) \rightarrow \int g(\theta) \pi(\theta|x) d\theta, \quad \text{as } N \rightarrow \infty$$

Theoretical basis: Strong Law of Large numbers: (here: $X_i = g(\theta^{(i)})$)

Strong Law of Large Numbers

Consider a sequence $X_1, X_2, \dots, X_N$ of independent and identically distribution (i.i.d.) random variables with expected value $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \dots = \mu$. The strong law of large numbers states that the sample average converges *almost surely* to the expected value

$$\frac{1}{N} \sum_{i=1}^N X_i \rightarrow \mu \quad \text{as } N \rightarrow \infty.$$

In words: the probability that the sample average converges to the expectation, as the number of samples goes to infinity, is equal to one.

Multivariate Monte Carlo

How to compute marginal distributions from joint distribution?

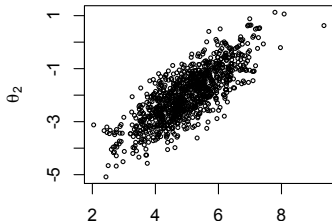
$$p(\theta_1|x) = \int_{\Theta_2} \pi(\theta_1, \theta_2|x) d\theta_2$$

- ▶ Suppose we have samples $(\theta_1^{(i)}, \theta_2^{(i)}) \sim \pi(\theta_1, \theta_2|x)$
- ▶ Integrating over θ_2 : $\Rightarrow$ discard the θ_2 part of the sample

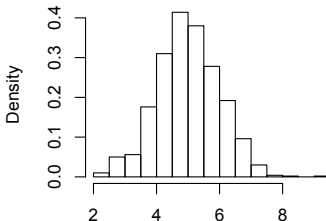
$$(\theta_1^{(i)}, \theta_2^{(i)}) \rightarrow \theta_1^{(i)}$$

- ▶ Construct histogram of $\theta_1^{(1)}, \dots, \theta_1^{(N)}$

Bivariate Posterior Sample



Estimated Marginal Distribution



Multivariate Monte Carlo

How to compute distributions of functions of parameters?

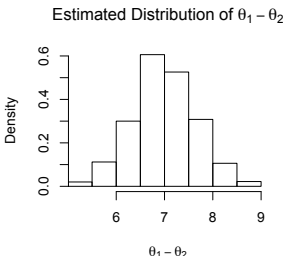
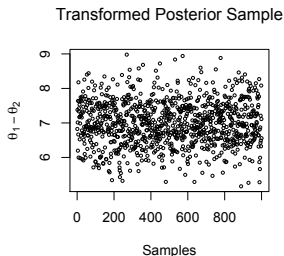
$$p(\theta_1 - \theta_2 | x) = \int_{\Phi_2} \pi(\theta_1, \theta_2 | x) \left| \frac{d(\theta_1, \theta_2)}{d(\phi_1, \phi_2)} \right| d\phi_2$$

where $\phi = (\phi_1, \phi_2)$, with $\phi_1 = \theta_1 - \theta_2$ and e.g. $\phi_2 = \theta_1 + \theta_2$

- ▶ Given samples $(\theta_1^{(i)}, \theta_2^{(i)}) \sim \pi(\theta_1, \theta_2 | x)$, compute

$$(\theta_1^{(i)}, \theta_2^{(i)}) \rightarrow \theta_1^{(i)} - \theta_2^{(i)} \quad \text{for } i = 1, \dots, N$$

- ▶ Construct histogram of $\theta_1^{(1)} - \theta_2^{(1)}, \dots, \theta_1^{(N)} - \theta_2^{(N)}$



Monte Carlo posterior predictive distributions

How to generate samples from $p(y|x)$?

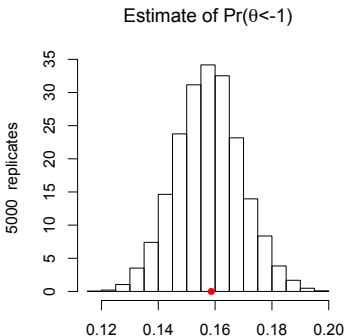
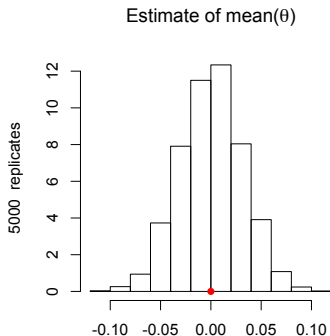
$$p(y|x) = \int_{\Theta} \pi(y|\theta)\pi(\theta|x)d\theta$$

By **inspection of formula**:

- ▶ Suppose: we already have samples $\theta^{(i)} \sim \pi(\theta|x)$
- ▶ For each $\theta^{(i)}$ we can generate $y^{(i)} \sim \pi(y|\theta^{(i)})$
- ▶ This gives us joint samples $(\theta^{(i)}, y^{(i)}) \sim \pi(y|\theta)\pi(\theta|x)$
- ▶ To obtain samples from $p(y|x)$, we “**integrate out**” θ (i.e. **discard the $\theta^{(i)}$ values**) to leave $y^{(i)} \sim p(y|x)$ only

Hugely simpler than calculating exact algebraic expression!

Monte Carlo error



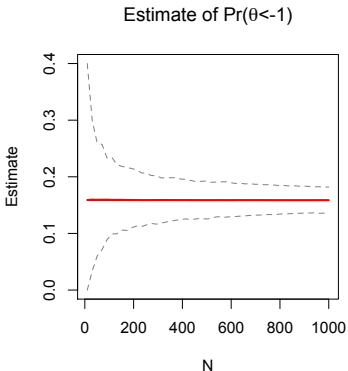
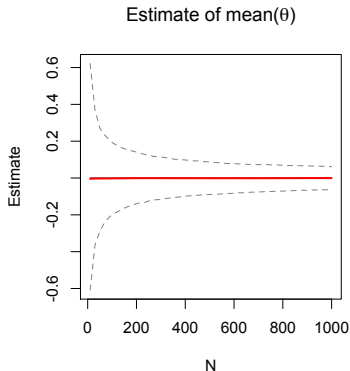
Estimating quantities with random samples gives random estimates
Estimate variability is known as Monte Carlo error.

$$N = 1000, \theta^{(i)} \sim N(0, 1)$$

Reporting estimates of posterior quantities is commonly accompanied by
(estimates of!) Monte Carlo error (recursion!)

(Not to be confused with posterior credible intervals ...)

Monte Carlo error changes with sample size, N



Increasing N reduces the Monte Carlo error

For greater precision of estimates, increase number of samples, N

Given a target precision, continue to sample until achieved

Equivalent concept to classical “sample size computations” to find significant experimental results

Bayesian inference and computation

1763 – 1984 Bayesian statistics is restricted to very simple analyses where exact integrations are possible

Almost useless as a general statistical method!

1984 – 2016 Complex integrations trivialised due to advent of cheap computing power and powerful Monte Carlo methods

Bayesian statistics now a mainstream statistical technique

2016 – ∞ New computational ML/AI tools further transform Bayesian modelling capabilities. New generalisations of Bayesian inference developed (“post-Bayesian inference”).

Field currently developing rapidly.

- ▶ Monte Carlo methods have made Bayesian inference tractable, even in very complex models.
- ▶ Bayesian development closely linked to computational capabilities
- ▶ This course: up to $\sim$ 2016